GANESHKUMAR JAGANATHAN
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PROFESSIONAL SUMMARY

Expert Cloud and Systems Engineer with 15+ years of comprehensive experience in UNIX/Linux system administration seamlessly integrated with advanced proficiency in Amazon Web Services (AWS). Specialized in crafting, deploying, and managing high-performing cloud architectures tailored to the unique demands of development, QA, staging, and production environments. Renowned for applying a robust skill set in AWS cloud services to achieve scalable, secure, and resilient systems. Exceptionally skilled in utilizing DevOps strategies, infrastructure automation, and chaos engineering to ensure peak system performance and reliability. A proven leader in merging deep technical acumen with innovative cloud solutions to drive operational excellence and strategic business outcomes in complex technical landscapes.

CORE COMPETENCIES

Cloud Solutions Architecture: Expert in designing and managing robust AWS cloud infrastructures, ensuring optimal alignment with business needs across various application environments.

Comprehensive AWS Expertise: Deep understanding of a wide range of AWS services including EC2, ELB, Auto-Scaling, S3, IAM, VPC, and more, enhancing system scalability and security.

Dynamic Resource Management: Effective in dynamically scaling AWS resources based on business demands, utilizing Auto Scaling, Lambda functions, and other AWS mechanisms to optimize costs and performance.

DevOps and Infrastructure Automation: Advanced proficiency in Continuous Integration (CI) and Continuous Delivery (CD) using tools such as Jenkins, GitLab, and AWS Code Pipeline.

Skilled in deploying Infrastructure as Code (IaC) using Terraform to streamline and automate configurations.
Chaos Engineering: Experienced in implementing chaos-engineering principles using Gremlin to proactively test and improve system resilience and disaster recovery procedures.
Kubernetes and Micro services: Proficient in setting up and managing AWS EKS clusters for deploying and scaling micro services in production environments.

TECHNICAL SKILLS:

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| Cloud Technology: | AWS |
| Infrastructure Automation: | Cloud Formation / Terraform |
| Configuration Automation: | Ansible |
| Continuous Integration: | Jenkins, GitLab and Code Pipeline |
| Analytical Tools: | Elastic Search Service and Splunk |
| Operating System: | Red Hat Enterprise Linux and Windows |
| Scripting: | Shell (BASH) and Python |
| Container Technology: | Kubernetes |
| Database: | RDS and Dynamo DB |

CERTIFICATIONS:
HashiCorp Certified: Terraform Associate
Microsoft Certified: Azure Fundamentals
AWS Certified Solutions Architect - Associate
Splunk Certified User
Gremlin Certified Chaos Engineering Practitioner

EDUCATION:
Master of Science in Information Technology (M.Sc. (IT))
Bachelor of Computer Application (BCA)

PROFESSIONAL EXPERIENCE:

**Cognizant Technology Solutions US Corp (Mar 2022 Till Date)**

Location : Remote (Dallas, TX)
Role : Lead Cloud Engineer

Project Overview :
The project modernized and improved the performance of the HealthCare portal, enabling faster processing times, enhanced security, and a better user experience for healthcare providers and patients. The modernization effort focused on scalability, automation, and efficiency, ensuring the portal can handle growing data volumes and evolving business needs.

Responsibilities :

Leading the design, development, and deployment of scalable, secure AWS cloud infrastructures, integrating advanced services such as S3, EC2, AWS Lambda, Step Functions, Load Balancers, Amazon Redshift, AWS Glue, AWS DMS, Amazon RDS, Athena, Dynamo DB, and more to support next-generation cloud platforms.
Developing AWS infrastructure from scratch using Terraform, defining custom Terraform modules, implementing advanced state management strategies (S3 + Dynamo DB backend), handling remote execution plans, and ensuring best practices for Infrastructure as Code (IaC) without relying on pre-existing templates. Architecting modular, reusable Terraform configurations to support multi-region and multi-account AWS environments.
Automating infrastructure provisioning and management, integrating Terraform with AWS Code Pipeline and Jenkins / GitLab CI/CD to enforce infrastructure validation, security scanning (Checkov), automated deployment, and compliance policies.
Provisioning, maintaining, and optimizing AWS services, including AWS Lambda and Step Functions for orchestration and automation, Amazon S3 for scalable storage, AWS Glue and AWS DMS for data processing and movement, AWS Direct Connect for hybrid cloud connectivity and AWS Marketplace Sandbox for data collaboration.
Automating configuration management and application deployment using Python and Ansible, developing custom Ansible roles and playbooks to manage EC2 configurations, package installations, security hardening, and operational automation.
Implementing security best practices and governance frameworks by enforcing IAM policies, AWS WAF integration, encryption standards, and compliance with regulatory frameworks.

Building and maintaining comprehensive monitoring solutions with AWS Cloud Watch, AWS Cloud Trail, and AWS Systems Manager, enabling real-time alerting, log aggregation, anomaly detection, and performance tuning for AWS workloads.
Establishing fully automated continuous integration pipelines using Jenkins, SonarQube, WhiteSource, and Frog to enable efficient code integration, security scanning, vulnerability detection, and artifact management.
Managing version control with Git/GitLab to support Terraform-based infrastructure changes, Ansible playbook automation, and application deployments, fostering collaborative development across global teams.
Designing and maintaining disaster recovery (DR) and high-availability (HA) strategies, implementing Terraform-driven backup policies (AWS Backup, RDS snapshots, cross-region S3 replication), failover mechanisms (Route 53 health checks, Multi-AZ deployments), and Ansible automation for DR testing to ensure business continuity in case of system failures.
Collaborating effectively with cross-functional teams, strategic partners, and global stakeholders, including cloud architects, DevOps engineers, security teams, data engineers, developers, and operations personnel to design, implement, and optimize AWS cloud infrastructure solutions.

Info stretch Corporation (Oct 2021 – Nov 2021)

Location : Remote (Austin, TX)
Role : Chaos Engineer

Project Overview :
This project focused on enhancing system resilience and reliability by integrating chaos management into the deployment pipeline for a leading financial institution. The goal was to proactively test and improve system stability under real-world failure scenarios, ensuring high availability and seamless user experience for financial services.

Responsibilities :

Collaborated with the Application team to analyze application architecture and identify critical areas suitable for chaos testing, enhancing system resilience.

Developed and maintained Jenkins pipelines to automate chaos testing, utilizing Gremlin APIs to streamline the process and ensure scalable deployments.

Worked closely with the Performance and Automation testing teams to integrate pre-chaos and post-chaos validation into the testing pipelines, facilitating comprehensive evaluations and improvements in system robustness.

Conducted chaos tests in partnership with the Application team, monitoring and analyzing outcomes to assess and refine system performance and stability metrics.

Designed and implemented robust monitoring solutions that tracked the real-time impact of chaos experiments, enabling immediate data-driven decision-making to optimize system reliability.

Led training sessions and workshops for new team members on chaos engineering principles and practices, promoting a culture of continuous learning and proactive testing across the organization.

Lifesize Inc (Jun 2018 – Oct 2021)

Location : Austin TX

Role : Sr. Cloud Engineer [First 3 months through Robert Half in LifeSize as Cloud Engineer]

Project Overview:

The project focused on modernizing and optimizing cloud infrastructure to enhance scalability, security, and performance for cloud-based contact center solutions. The initiative aimed to streamline deployment processes, improve system reliability, and support business growth through cloud automation and infrastructure enhancements.

Responsibilities :

Design and deployment of scalable, high-availability AWS cloud infrastructures utilizing EC2, Auto Scaling, Load Balancers, and RDS to optimize application and database performance.
Automated infrastructure tasks using Jenkins and GitHub, significantly enhancing deployment efficiency and system reliability.
Administered AWS EKS clusters and managed Kubernetes applications using Helm, ensuring stable and reproducible builds across environments.
Utilized Prometheus for proactive monitoring and management of Kubernetes clusters, improving system visibility and performance.
Managed robust AWS storage solutions, including S3 and Elastic Block Storage, to ensure data integrity and availability.
Handled multiple database platforms such as MySQL, MSSQL, Aurora RDS, Dynamo DB, and Radis, optimizing configurations for performance and scalability.
Implemented infrastructure as code using Cloud Formation and Terraform to streamline cloud operations and maintain consistency across environments.
Deployed various monitoring tools including Cloud Watch, LogicMonitor, and OpsGenie to maintain system health and quickly address potential issues.
Collaborated effectively with business, development, and DevOps teams, aligning technical solutions with strategic business objectives and ensuring seamless project execution.

Cognizant Technology Solutions US Corp (Mar 2013 – Jun 2018)

Duration : Mar 2013 – Jun 2018
Location : Austin, TX
Role : Cloud Technology Specialist
Responsibilities :

Launched and configured Amazon EC2 instances using AWS (Linux/Ubuntu), tailoring environments to meet specific application requirements.
Utilized AWS services such as EC2, Auto-Scaling, and VPC to construct secure, scalable systems capable of handling dynamic load variations.

Architected and implemented comprehensive cloud solutions on AWS, focusing on high availability and security.
Developed 3-tier architecture systems integrating Web, Application, and Database servers, using AWS Elastic Load Balancer (ELB) and Route 53 for DNS management.
Designed roles and managed access using AWS Identity and Access Management (IAM) to enhance organizational security.
Enhanced network security by configuring AWS security groups, network ACLs, Internet Gateways, and Elastic IPs.
Set up monitoring with AWS CloudWatch to track server performance, CPU utilization, and disk usage, enabling proactive performance optimization.
Supported enterprise Java applications on Linux-based systems, integrating Oracle and MySQL databases.
Ensured application security compliance for internal audits, managing user and application account audits in accordance with standard guidelines.
Played a key role in developing the CI/CD pipeline utilizing Jenkins, Maven, GitHub, and Ansible on AWS.
Configured Jenkins on Linux for master-slave operations, facilitating multiple parallel builds.
Managed web applications and environment configurations using Ansible, automating tasks related to files, users, and package management.
Authored UNIX shell scripts to automate jobs, scheduling cron jobs for daily and nightly operations.
Automated and streamlined nightly builds and daily tasks through scripting and Jenkins, significantly improving efficiency.
Coordinated with developers to apply best practices for GIT branching and labeling/naming conventions to enhance source control management.
Integrated GIT with Jenkins to automate code checkout processes, enhancing continuous integration workflows.
Provided monitoring, management, and troubleshooting support for systems and services hosted on cloud resources, ensuring high uptime and reliability.

COGNIZANT TECHNOLOGY SOLUTIONS (Jun 2010 – Mar 2013)

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| Location | : | Chennai, India |
| Role | : | Offshore Technical Lead |
| Responsibilities | : | |

Developed, installed, administered, and maintained Linux-based infrastructure, ensuring robust performance and high availability.
Built and configured Linux servers, setting up both physical and virtual network environments to support organizational IT needs.
Configured and managed application high availability using Red Hat Cluster Suite, enhancing system resilience and uptime.
Managed file systems using Logical Volume Manager (LVM), optimizing storage solutions and ensuring data integrity.
Conducted regular upgrades of the Red Hat Linux operating system to maintain system security and functionality.
Automated server and cluster health checks to proactively address potential issues, improving system reliability and performance.
Monitored system performance including virtual memory, system events, swapping, disk utilization, and CPU usage through custom scripts.
Created and maintained shell scripts to automate repetitive system administration tasks, reducing manual effort and increasing operational efficiency.